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Experiment No. : 9

Objective: Determination of (i) Voltage ratio (ii) polarity of a single phase transformer.

Apparatus Required:

1. 1-phase Transformer
2. 3 Voltmeters.
3. Connecting wires

Circuit Diagram:

1. Voltage ratio

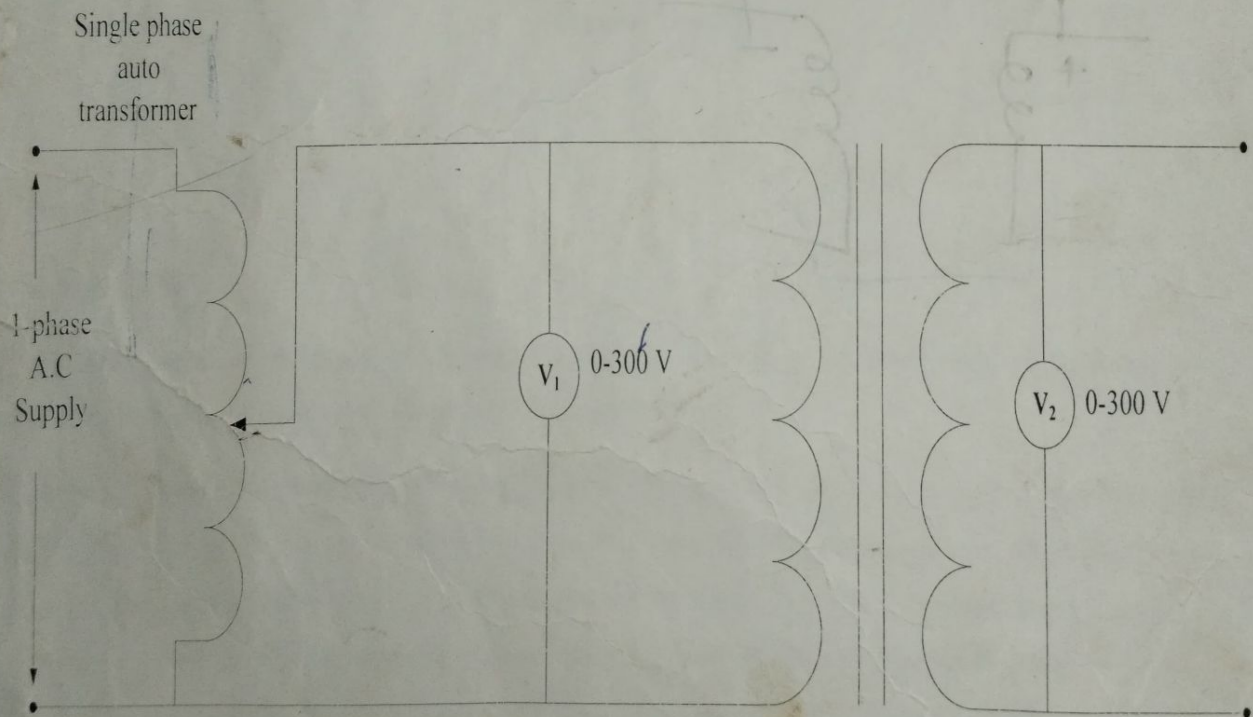


Figure: (a)

2. Polarity test

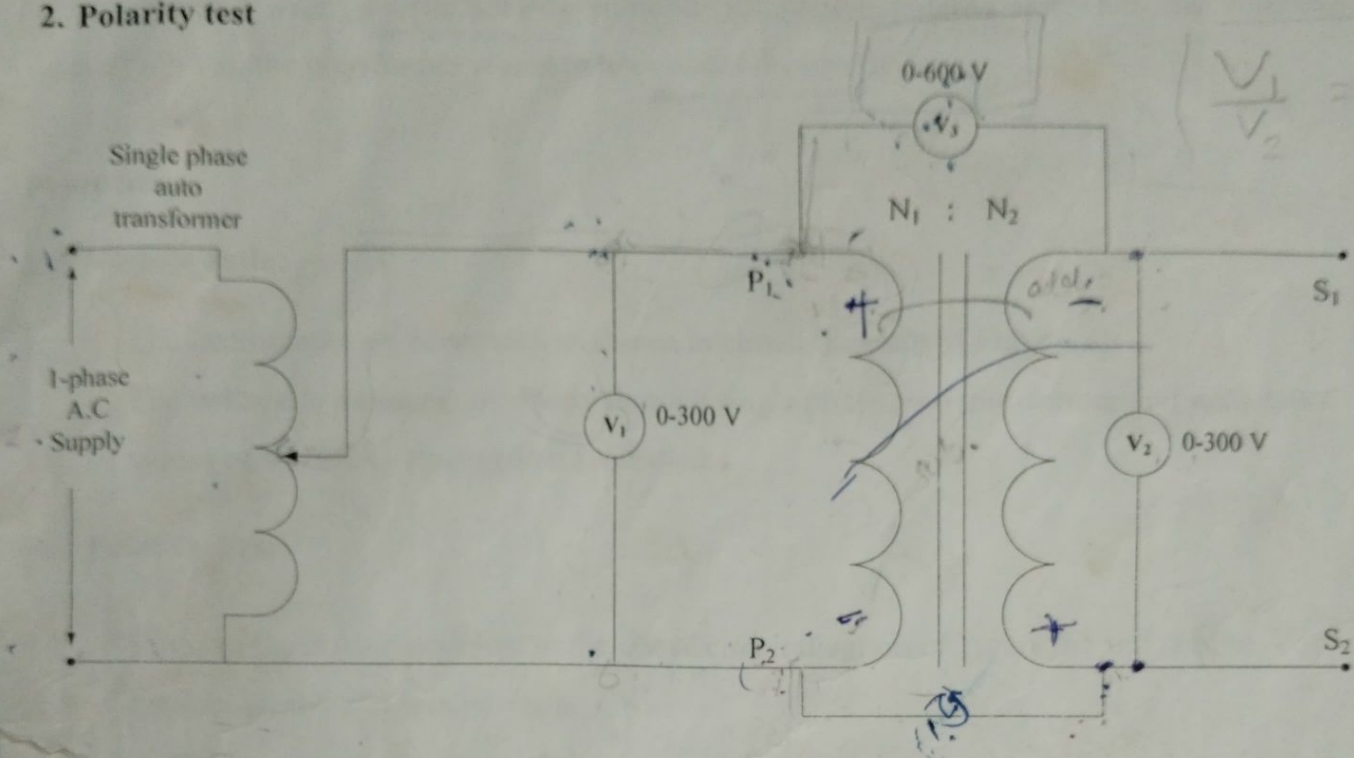


Figure: (b)

Theory:

Voltage Ratio:

The turn's ratio of a single phase transformer is defined as ratio of number of turns of the secondary coil to the number of turns of the primary coil.

If N_2 is the number of turns of secondary coil (winding) and N_1 is the number of turns of primary coil (winding) then the turns ratio $K = N_2 / N_1$. As the frequency of the alternating current remains the same while flowing through the transformer then the voltage per unit turn will remain the same for both primary and secondary coils so $V_1 / N_1 = V_2 / N_2$ or $N_2 / N_1 = V_2 / V_1$.

Thus the turns ratio, $K = N_2 / N_1 = V_2 / V_1$ where V_1 and V_2 are the corresponding voltages across the primary and secondary coils respectively.

Polarity Test: When the voltmeter reads the difference that is $(V_1 - V_2)$, the transformer is said to be connected with opposite polarity known as Subtractive polarity and when the voltmeter reads $(V_1 + V_2)$, the transformer is said to have additive polarity.

Procedure:-

For Voltage ratio:

1. The instruments are connected as shown in circuit diagram of Figure (a)
2. The voltage is increased gradually through single phase auto transformer and successive values of V_1 and V_2 observed and recorded.

For Polarity Test:

1. Connect the circuit as shown in the above circuit diagram of Figure (b) and set the autotransformer to zero position.
2. Switch on the single phase supply
3. Records the values of the voltages as shown by the voltmeter V_1 , V_2 and V_3 .
4. If the reading of the V_3 shows the addition of the value of V_1 and V_2 that is $V_3 = V_1 + V_2$ the transformer is said to be connected in additive polarity.
5. If the reading of the V_3 is the subtraction of the readings of V_1 and V_2 , then the transformer is said to be connected in subtractive or negative polarity.

Observation Table:

S.NO.	V_1	V_2	$K = V_2/V_1$
1.			
2.			
3.			
4.			
5.			

S.NO.	V_1	V_2	$V_1 - V_2$	$V_1 \neq V_2$	V_3	Polarity type
						+
						-

RESULTS AND CONCLUSION:

PRECAUTIONS:

- 1) Do not put on the supply until the circuit is checked by concerned teacher.
- 2) Do not touch any live part of circuit.
- 3) Be careful for primary & secondary winding rated current.
- 4) Switch ON the supply after verification of circuit connections by lab incharge /faculty.